



Ghana communication technology University

Department of Information Technology

BSc. Information Technology

STUDENT ACADEMIC PROGRESS TRACKING SYSTEM

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BACKGROUND TO THE STUDY

- Students manage several courses, each with its own assessments and deadlines.
- Progress is tracked across notebooks, calculators and calendars that do not communicate.
- Nothing connects study hours to the marks eventually received.
- Zimmerman (2002): metacognitive awareness correlates strongly with achievement.
- A technology university is well placed to solve this for its own students.

PROBLEM STATEMENT

Students have no integrated means of monitoring academic progress:

- Fragmented data — marks, deadlines and notes held in unconnected places.
- No visualisation — a bare mark hides trends and improvement.
- Effort disconnected from outcome — study time is never recorded.
- Unstructured goals — objectives are never broken into trackable steps.

Generic tools have no concept of credit hours or assessment weighting.

AIM & OBJECTIVES

Aim

- To design, develop and evaluate a web-based Student Academic Progress Tracking System, with an account for each student, deployed on XAMPP.

Objectives

- Analyse student tracking needs.
- Design an intuitive, responsive interface.
- Develop the core academic modules.
- Implement a relational data layer with access control.
- Provide administrator enrolment of students.
- Test for correctness, usability, security and performance.

SIGNIFICANCE OF THE STUDY

- Practical — deployable by a department on one machine, at no cost and without internet.
- Privacy — records never leave the institution's own hardware.
- Academic — operationalises Self-Regulated Learning theory in software.
- Technical — a complete multi-user system on the free XAMPP stack.
- For the Faculty — a documented reference implementation for later projects.

LITERATURE REVIEW

Theoretical framework — Self-Regulated Learning (Zimmerman, 2002)

- Forethought → goals; Performance → study tracker; Reflection → grade analysis.

Feedback and assessment

- Black and Wiliam (1998); Hattie and Timperley (2007) — feedback works when it relates performance to a goal.

Systems reviewed

- MyStudyLife, Todoist, grade calculators, institutional LMS and SIS, Course Signals.

Research gap

- None combines integration, visualisation, academic specificity and zero cost.

METHODOLOGY

Research design

- System development — the question is answered by building an artefact and evaluating it.

Data collection

- Interviews with students and lecturers; observation of record-keeping; review of existing tools.

Development model — Waterfall (Royce, 1970)

- Requirements → Design → Implementation → Testing → Deployment.
- Chosen because requirements were stable and each phase yields a documented deliverable.

SYSTEMS ANALYSIS

Actors

- Student — all academic tracking. Administrator — enrolls students, but cannot read marks.

Functional requirements (13)

- Authentication, enrolment, record separation, courses, weighted assessment and GPA, study logging, goals, flashcards, vault, charts, persistence, backup.

Non-functional requirements (8)

- Responsive 360–1920 px; interactive within 3 seconds; learnable in 10 minutes; hashed passwords; reliability; offline operation; maintainability; auditability.

SYSTEM DESIGN

Three-tier architecture across client and server

- Presentation — HTML5, CSS3. Logic — JavaScript in the browser, PHP 8 on the server. Data — MySQL.

Database

- Eight tables with primary keys, foreign keys and check constraints; four views reproducing the grade and GPA calculations in SQL.

Security

- bcrypt hashing (cost 12); HttpOnly SameSite sessions; CSRF tokens; PDO prepared statements; every query filtered by the account in the session.

SYSTEM IMPLEMENTATION

- HTML5, CSS3 and JavaScript on the client; PHP 8 and MySQL on the server; hosted on XAMPP at <http://localhost/sapts/>.
- Eight student modules plus an administrator enrolment console.
- Tested: correct, wrong and unknown sign-in; forced first-password change; a course added in the interface arriving correctly in the database.
- A second student could not see the first student's records; administrator routes return 403 to a student.

CONCLUSIONS AND RECOMMENDATIONS

Conclusions

- Grading, study tracking, revision and goal setting are consolidated into one interface, closing the gap found in the literature.
- Records are held under each student's own account and are unreadable to others.
- A multi-user academic system can run entirely on the free XAMPP stack.

Recommendations

- Serve over HTTPS before any network deployment.
- Import marks automatically from the institutional LMS.
- Add a lecturer role for aggregate, anonymised course performance.

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The full reference list appears in the project report.